

Full assumption inventory: scope and navigation

1,747 entries | 39 groups | exact declared values, including missing inputs and duplicate records

Input family	Fuel / calculation affected	PDF pages
Agriculture	Diesel 50 ppm	2–4
Aviation	Jet A1	5–10
Generation	Diesel for power	11–16
Industrial	Diesel 50 ppm	17–20
Macro	Shared drivers / conversions; use depends on module	21–32
Marine	Marine diesel and supplementary fuel oil	33–35
Supply	Domestic petrol, diesel and jet production	36–46
Vehicles	Petrol and road diesel	47–55
Calculation conventions	Embedded constants and implementation rules	56

Scope: every registered scalar and CSV observation in the current 2026 draft. Declared inputs can be unused; inclusion is not validation.

M, available imports and WS5 capacity/access inputs remain to build. Embedded conventions follow the register inventory; EXC-CODE remains open.

Assumption inventory: Agriculture • Base Year Volume

2026 draft | Diesel 50 ppm | REG-AGRICULTURE-BASE_YEAR_VOLUME | 1/1

Units: litres per year (diesel)

Parameter / scenario	Coverage	Declared value
value	ZAF	0.7e9
year	ZAF	2024
value	BWA, LSO, NAM, SWZ	Missing (null)
year	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Agriculture • Elasticity

2026 draft | Diesel 50 ppm | REG-AGRICULTURE-ELASTICITY | 1/1

Units: dimensionless

Parameter / scenario	Coverage	Declared value
value	ZAF	0.2
value	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Agriculture • Product

2026 draft | Diesel 50 ppm | REG-AGRICULTURE-PRODUCT | 1/1

Units: product code

Parameter / scenario	Coverage	Declared value
default	Shared	diesel_50ppm

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Aviation • Regression

2026 draft | Jet A1 | REG-AVIATION-REGRESSION | 1/1

Units: litres per year (response); ZAR/capita and pax (drivers)

Parameter / scenario	Coverage	Declared value
intercept	ZAF	-14746408795.767437
coefficients.gdp_per_capita	ZAF	196334.9338396717
coefficients.passenger_departures	ZAF	79.78786549099274
r_squared	ZAF	0.94602
observations	ZAF	6
value	BWA, LSO, NAM, SWZ	Missing (null)

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Aviation • Passenger Departures

2026 draft | Jet A1 | REG-AVIATION-PASSENGER_DEPARTURES | 1/3

Units: pax per year

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2017	20836846.0	—	—
ZAF	2018	21118264.0	—	—
ZAF	2019	20924465.0	—	—
ZAF	2020	4569737.0	—	—
ZAF	2021	10538202.0	—	—
ZAF	2022	15813215.0	—	—
ZAF	2023	17730490.28571429	—	—
ZAF	2024	17000000.0	—	—
ZAF	2025	19000000.0	—	—
ZAF	2026	21000000.000000004	—	—
ZAF	2027	22366222.610112537	—	—
ZAF	2028	23832226.837557185	—	—
ZAF	2029	25394320.978455253	—	—
ZAF	2030	27058803.289861277	—	—
ZAF	2031	28832384.850951277	—	—
ZAF	2032	30722216.621636357	—	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Aviation • Passenger Departures

2026 draft | Jet A1 | REG-AVIATION-PASSENGER_DEPARTURES | 2/3

Units: pax per year

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2033	32735918.27474543	—	—
ZAF	2034	34881608.91802647	—	—
ZAF	2035	37167939.829835296	—	—
ZAF	2036	39604129.34049995	—	—
ZAF	2037	42199999.99999999	—	—
ZAF	2038	44966018.18181817	—	—
ZAF	2039	47913336.28264462	—	—
ZAF	2040	51053837.68807978	—	—
ZAF	2041	54400184.68563483	—	—
ZAF	2042	57965869.51821144	—	—
ZAF	2043	61765268.78390512	—	—
ZAF	2044	65813701.40146836	—	—
ZAF	2045	70127490.37514642	—	—
ZAF	2046	74724028.60791738	—	—
ZAF	2047	79621849.02849089	—	—
ZAF	2048	84840699.31481288	—	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Aviation • Passenger Departures

2026 draft | Jet A1 | REG-AVIATION-PASSENGER_DEPARTURES | 3/3

Units: pax per year

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2049	90401621.51535653	—	—
ZAF	2050	96327036.88922672	—	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Aviation • Product Split

2026 draft | Jet A1 | REG-AVIATION-PRODUCT_SPLIT | 1/1

Units: fraction (sums to 1 per country)

Parameter / scenario	Coverage	Declared value
jet_a1	ZAF, BWA, LSO, NAM, SWZ	1.0

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Aviation • Seasonality

2026 draft | Jet A1 | REG-AVIATION-SEASONALITY | 1/1

Units: not recorded

Parameter / scenario	Coverage	Declared value
value	ZAF	aviation_za
value	BWA	aviation_bw
value	LSO	aviation_ls
value	NAM	aviation_na
value	SWZ	aviation_sz

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Generation • Ocgt Capacity

2026 draft | Diesel for power | REG-GENERATION-OCGT_CAPACITY | 1/1

Units: MW

Parameter / scenario	Coverage	Declared value
stations.ankerlig	ZAF	1327
stations.gourikwa	ZAF	740
stations.avon	ZAF	670
stations.dedisa	ZAF	335
stations.new_1	ZAF	1000
stations.new_2	ZAF	2000
value	BWA, LSO, NAM, SWZ	Missing (null)

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Generation • Load Factor

2026 draft | Diesel for power | REG-GENERATION-LOAD_FACTOR | 1/1

Units: fraction

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2022	—	0.45	0.45
ZAF	2023	—	0.5603	0.5603
ZAF	2024	—	0.5456	0.5456
ZAF	2025	—	0.626	0.626
ZAF	2026	—	0.7	0.14133780969896842
ZAF	2027	—	0.71	0.1461268140643498
ZAF	2028	—	0.7	0.15091581842973117
ZAF	2029	—	0.7	0.15570482279511255
ZAF	2030–2050	—	0.7	0.16049382716049385

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Generation • Load Shedding

2026 draft | Diesel for power | REG-GENERATION-LOAD_SHEDDING | 1/1

Units: GWh

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2024	—	31578	34
ZAF	2025	—	29341	25
ZAF	2026	—	39587	84
ZAF	2027	—	41103	235
ZAF	2028–2050	—	40743	323

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Generation • Efficiency

2026 draft | Diesel for power | REG-GENERATION-EFFICIENCY | 1/1

Units: fraction (thermal efficiency)

Parameter / scenario	Coverage	Declared value
value	ZAF	0.4
value	BWA, LSO, NAM, SWZ	Missing (null)

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Generation • Mj Per Litre

2026 draft | Diesel for power | REG-GENERATION-MJ_PER_LITRE | 1/1

Units: MJ per litre (diesel)

Parameter / scenario	Coverage	Declared value
value	Shared	36.9

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Generation • Operational Days

2026 draft | Diesel for power | REG-GENERATION-OPERATIONAL_DAYS | 1/1

Units: days per year

Parameter / scenario	Coverage	Declared value
value	Shared	365

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Industrial • Base Year Volume

2026 draft | Diesel 50 ppm | REG-INDUSTRIAL-BASE_YEAR_VOLUME | 1/1

Units: litres per year (diesel)

Parameter / scenario	Coverage	Declared value
value	ZAF	2.5e9
year	ZAF	2024
value	BWA, LSO, NAM, SWZ	Missing (null)
year	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Industrial • Elasticity

2026 draft | Diesel 50 ppm | REG-INDUSTRIAL-ELASTICITY | 1/1

Units: dimensionless (response of diesel demand to GDP/capita)

Parameter / scenario	Coverage	Declared value
value	ZAF	1.0
value	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Industrial • Product

2026 draft | Diesel 50 ppm | REG-INDUSTRIAL-PRODUCT | 1/1

Units: product code

Parameter / scenario	Coverage	Declared value
default	Shared	diesel_50ppm

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Industrial • Seasonality

2026 draft | Diesel 50 ppm | REG-INDUSTRIAL-SEASONALITY | 1/1

Units: not recorded

Parameter / scenario	Coverage	Declared value
value	ZAF, BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Macro • Gdp

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-GDP | 1/3

Units: ZAR (constant 2015)

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2005	3411409882751.5645	—	—
ZAF	2006–2007	3602578678405.883	—	—
ZAF	2008	3916816252461.046	—	—
ZAF	2009	3856572134210.3213	—	—
ZAF	2010	3973801603311.196	—	—
ZAF	2011	4099713743523.5083	—	—
ZAF	2012	4197952411913.467	—	—
ZAF	2013	4302291176143.1406	—	—
ZAF	2014	4363118106800.599	—	—
ZAF	2015	4420792517414.472	—	—
ZAF	2016	4450170996114.515	—	—
ZAF	2017	4501701615484.83	—	—
ZAF	2018	4568669710783.134	—	—
ZAF	2019	4573834760819.281	—	—
ZAF	2020	4279646860386.556	—	—
ZAF	2021	4513043919695.861	—	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Gdp

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-GDP | 2/3

Units: ZAR (constant 2015)

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2022	4599261389164.597	—	—
ZAF	2023	4638499639146.34	—	—
ZAF	2024	4659294000000	—	—
ZAF	2025	4679355000000	—	—
ZAF	2026	—	4754224680000	4726148550000
ZAF	2027	—	4830292274880	4773410035500
ZAF	2028	—	4907576951278.08	4821144135855
ZAF	2029	—	4986098182498.529	4869355577213.55
ZAF	2030	—	5065875753418.506	4918049132985.686
ZAF	2031	—	5146929765473.202	4967229624315.542
ZAF	2032	—	5229280641720.773	5016901920558.697
ZAF	2033	—	5312949131988.306	5067070939764.284
ZAF	2034	—	5397956318100.118	5117741649161.927
ZAF	2035	—	5484323619189.72	5168919065653.546
ZAF	2036	—	5572072797096.755	5220608256310.081
ZAF	2037	—	5661225961850.303	5272814338873.182

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Gdp

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-GDP | 3/3

Units: ZAR (constant 2015)

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2038	—	5751805577239.907	5325542482261.913
ZAF	2039	—	5843834466475.746	5378797907084.532
ZAF	2040	—	5937335817939.358	5432585886155.378
ZAF	2041	—	6032333191026.389	5486911745016.932
ZAF	2042	—	6128850522082.811	5541780862467.101
ZAF	2043	—	6226912130436.136	5597198671091.771
ZAF	2044	—	6326542724523.114	5653170657802.689
ZAF	2045	—	6427767408115.484	5709702364380.717
ZAF	2046	—	6530611686645.332	5766799388024.524
ZAF	2047	—	6635101473631.657	5824467381904.77
ZAF	2048	—	6741263097209.764	5882712055723.817
ZAF	2049	—	6849123306765.12	5941539176281.056
ZAF	2050	—	6958709279673.362	6000954568043.866

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Gdp Per Capita

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-GDP_PER_CAPITA | 1/3

Units: ZAR (constant 2015) per capita

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2005	71248.2767340624	—	—
ZAF	2006	74296.11262368012	—	—
ZAF	2007	77274.26804462152	—	—
ZAF	2008	78683.36274159448	—	—
ZAF	2009	76402.54256348967	—	—
ZAF	2010	77587.60106413947	—	—
ZAF	2011	78834.95005665856	—	—
ZAF	2012	79457.52667707803	—	—
ZAF	2013	80136.36744644346	—	—
ZAF	2014	79992.36191342783	—	—
ZAF	2015	79117.19955519178	—	—
ZAF	2016	78872.59198582664	—	—
ZAF	2017	79477.49731131857	—	—
ZAF	2018	79677.34204766274	—	—
ZAF	2019	78741.03379521101	—	—
ZAF	2020	72780.72469268832	—	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Gdp Per Capita

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-GDP_PER_CAPITA | 2/3

Units: ZAR (constant 2015) per capita

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2021	75987.07810801023	—	—
ZAF	2022	76790.1650122451	—	—
ZAF	2023	76777.92621036293	—	—
ZAF	2024	76356.55727959426	—	—
ZAF	2025	75873.5403538539	—	—
ZAF	2026	—	76305.62080219547	75854.99705730061
ZAF	2027	—	76770.1127412939	75866.05648927581
ZAF	2028	—	77267.03031188184	75906.194397226
ZAF	2029	—	77793.85014588524	75972.41454458526
ZAF	2030	—	78347.23662423984	76060.99673716872
ZAF	2031	—	78904.55961677714	76149.68221467329
ZAF	2032	—	79465.847125887	76238.47109752694
ZAF	2033	—	80031.12735315424	76327.3635062981
ZAF	2034	—	80600.42870077565	76416.35956169573
ZAF	2035	—	81173.77977298702	76505.45938456958
ZAF	2036	—	81751.20937750032	76594.66309591026

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Gdp Per Capita

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-GDP_PER_CAPITA | 3/3

Units: ZAR (constant 2015) per capita

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2037	—	82332.74652695116	76683.97081684948
ZAF	2038	—	82918.42044035652	76773.38266866018
ZAF	2039	—	83508.26054458284	76862.89877275671
ZAF	2040	—	84102.2964758245	76952.519250695
ZAF	2041	—	84700.55808109297	77042.24422417265
ZAF	2042	—	85303.07541971635	77132.07381502923
ZAF	2043	—	85909.87876484981	77222.00814524635
ZAF	2044	—	86520.99860499654	77312.04733694781
ZAF	2045	—	87136.46564553965	77402.19151239986
ZAF	2046	—	87756.310810285	77492.44079401127
ZAF	2047	—	88380.5652430149	77582.79530433354
ZAF	2048	—	89009.26030905296	77673.2551660611
ZAF	2049	—	89642.42759683992	77763.82050203138
ZAF	2050	—	90280.0989195209	77854.49143522508

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Gdp Growth

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-GDP_GROWTH | 1/1

Units: fraction per year (real GDP growth)

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2020	—	-0.06431974826743181	—
ZAF	2021	—	0.049	0.049
ZAF	2022	—	0.019	0.016
ZAF	2023	—	0.009999999999999998	0.004999999999999999
ZAF	2024	—	0.015	0.015
ZAF	2025	—	0.017	0.017
ZAF	2026–2050	—	0.016	0.009999999999999998

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Population

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-POPULATION | 1/3

Units: persons

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2005	47880595	—	—
ZAF	2006	48489464	—	—
ZAF	2007	49119766	—	—
ZAF	2008	49779472	—	—
ZAF	2009	50477013	—	—
ZAF	2010	51216967	—	—
ZAF	2011	52003759	—	—
ZAF	2012	52832659	—	—
ZAF	2013	53687125	—	—
ZAF	2014	54544184	—	—
ZAF	2015	55876504	—	—
ZAF	2016	56422274	—	—
ZAF	2017	56641210	—	—
ZAF	2018	57339635	—	—
ZAF	2019	58087055	—	—
ZAF	2020	58801927	—	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Population

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-POPULATION | 2/3

Units: persons

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2021	59392255	—	—
ZAF	2022	59893886	—	—
ZAF	2023	60414495	—	—
ZAF	2024	61020221	—	—
ZAF	2025	61673081	—	—
ZAF	2026	62305039	—	—
ZAF	2027	62918916	—	—
ZAF	2028	63514502	—	—
ZAF	2029	64093732	—	—
ZAF	2030	64659278	—	—
ZAF	2031	65229814.227096096	—	—
ZAF	2032	65805384.71372149	—	—
ZAF	2033	66386033.88083984	—	—
ZAF	2034	66971806.541373536	—	—
ZAF	2035	67562747.90366225	—	—
ZAF	2036	68158903.57495199	—	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Population

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-POPULATION | 3/3

Units: persons

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2037	68760319.56491494	—	—
ZAF	2038	69367042.28920037	—	—
ZAF	2039	69979118.57301684	—	—
ZAF	2040	70596595.65474607	—	—
ZAF	2041	71219521.1895887	—	—
ZAF	2042	71847943.25324209	—	—
ZAF	2043	72481910.34561078	—	—
ZAF	2044	73121471.39454953	—	—
ZAF	2045	73766675.75963944	—	—
ZAF	2046	74417573.23599738	—	—
ZAF	2047	75074214.05811904	—	—
ZAF	2048	75736648.90375595	—	—
ZAF	2049	76404928.89782657	—	—
ZAF	2050	77079105.61636202	—	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Macro • Fx Zar Usd

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-FX_ZAR_USD | 1/1

Units: ZAR per USD

Parameter / scenario	Coverage	Declared value
value	Shared	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Macro • Crude Oil Litres Per Barrel

2026 draft | Shared drivers / conversions; use depends on module | REG-MACRO-CRUDE_OIL_LITRES_PER_BARREL | 1/1

Units: litres per barrel

Parameter / scenario	Coverage	Declared value
value	Shared	158.987

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Marine • Base Year Volume

2026 draft | Marine diesel and supplementary fuel oil | REG-MARINE-BASE_YEAR_VOLUME | 1/1

Units: litres per year (per port)

Parameter / scenario	Coverage	Declared value
by_port.durban.country	Shared	ZAF
by_port.durban.value	Shared	1.5e9
by_port.durban.year	Shared	2024
by_port.cape_town.country	Shared	ZAF
by_port.cape_town.value	Shared	0.5e9
by_port.cape_town.year	Shared	2024
by_port.saldanha.country	Shared	ZAF
by_port.saldanha.value	Shared	0.2e9
by_port.saldanha.year	Shared	2024
by_port.walvis_bay.country	Shared	NAM
by_port.walvis_bay.value	Shared	0.25e9
by_port.walvis_bay.year	Shared	2024
by_port.luderitz.country	Shared	NAM
by_port.luderitz.value	Shared	0.05e9
by_port.luderitz.year	Shared	2024

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Marine • Elasticity

2026 draft | Marine diesel and supplementary fuel oil | REG-MARINE-ELASTICITY | 1/1

Units: dimensionless

Parameter / scenario	Coverage	Declared value
value	ZAF, NAM	0.4
value	BWA, LSO, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Marine • Product Split

2026 draft | Marine diesel and supplementary fuel oil | REG-MARINE-PRODUCT_SPLIT | 1/1

Units: fraction (sums to 1)

Parameter / scenario	Coverage	Declared value
default.fuel_oil	Shared	0.65
default.diesel_500ppm	Shared	0.35

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Supply • Refinery Capacity

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_CAPACITY | 1/1

Units: kbpd (thousand barrels per day)

Parameter / scenario	Coverage	Declared value
enref	ZAF	120
sapref	ZAF	180
natref	ZAF	108.5
sasol	ZAF	75
astron	ZAF	100
petrosa	ZAF	45
value	BWA, LSO, NAM, SWZ	Missing (null)

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Supply • Refinery Product Mix

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_PRODUCT_MIX | 1/2

Units: fraction

Parameter / scenario	Coverage	Declared value
enref.gasoline	ZAF	0.45
enref.diesel	ZAF	0.25
enref.jet_a1	ZAF	0.0
sapref.gasoline	ZAF	0.45
sapref.diesel	ZAF	0.25
sapref.jet_a1	ZAF	0.1
natref.gasoline	ZAF	0.45
natref.diesel	ZAF	0.25
natref.jet_a1	ZAF	0.1
sasol.gasoline	ZAF	0.85
sasol.diesel	ZAF	0.15
sasol.jet_a1	ZAF	0.0
astron.gasoline	ZAF	0.45
astron.diesel	ZAF	0.25
astron.jet_a1	ZAF	0.1
petrosa.gasoline	ZAF	0.85

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Supply • Refinery Product Mix

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_PRODUCT_MIX | 2/2

Units: fraction

Parameter / scenario	Coverage	Declared value
petrosa.diesel	ZAF	0.15
petrosa.jet_a1	ZAF	0.0
value	BWA, LSO, NAM, SWZ	Missing (null)

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Supply • Refinery Utilisation

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_UTILISATION | 1/7

Units: fraction

Country / asset	Year(s)	Shared	High demand: raw alternatives	Low
ZAF:astron	2022	—	0 [x2]	0
ZAF:astron	2023	—	0.5 [x2]	0.5
ZAF:astron	2024–2050	—	0.9; 0.8	0.8
ZAF:enref	2008	—	725799000; 8.893333333333333	—
ZAF:enref	2009	—	666873000; 7.2799999999999985	—
ZAF:enref	2010	—	698802000; 8.147499999999999	—
ZAF:enref	2011	—	754153000; 9.861666666666668	—
ZAF:enref	2012	—	766291000; 11.434166666666664	—
ZAF:enref	2013	—	781418000; 12.739166666666664	—
ZAF:enref	2014	—	830645000; 13.695833333333335	—
ZAF:enref	2015	—	805722000; 12.244166666666667	—
ZAF:enref	2016	—	823300000; 12.392916666666665	—
ZAF:enref	2017	—	895150000; 13.416153846153843	—
ZAF:enref	2018	—	946454000; 15.139999999999999	—
ZAF:enref	2019	—	923684000; 15.421249999999999	—
ZAF:enref	2020	—	815622000; 14.406666666666666	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Supply • Refinery Utilisation

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_UTILISATION | 2/7

Units: fraction

Country / asset	Year(s)	Shared	High demand: raw alternatives	Low
ZAF:enref	2021	—	852442000; 17.56	—
ZAF:enref	2022	—	0 [x2]; 0.45 [x2]; 993897000; 22.68	0
ZAF:enref	2023	—	0 [x2]; 0.5603 [x2]; 989792000; 22.79	0
ZAF:enref	2024	—	0 [x2]; 0.5456 [x2]; 0.02; 0.15; 1141616000; 22.2	0
ZAF:enref	2025	—	0 [x2]; 0.626 [x2]; 0.01; 0.15; 1136000282.275164; 21.59	0
ZAF:enref	2026	—	0 [x2]; 0.14133780969896842; 0.7; 0.02; 0.15	0
ZAF:enref	2027	—	0 [x2]; 0.1461268140643498; 0.71; 0.03; 0.15	0
ZAF:enref	2028	—	0 [x2]; 0.15091581842973117; 0.7; 0.04; 0.15	0
ZAF:enref	2029	—	0 [x2]; 0.15570482279511255; 0.7; 0.04; 0.15	0
ZAF:enref	2030	—	0 [x2]; 0.16049382716049385; 0.7; 0.04; 0.15	0
ZAF:enref	2031–2035	—	0 [x2]; 0.16049382716049385; 0.7; 0.04; 0.18	0
ZAF:enref	2036–2040	—	0 [x2]; 0.16049382716049385; 0.7; 0.04; 0.2	0
ZAF:enref	2041–2050	—	0 [x2]; 0.16049382716049385; 0.7 [x2]; 0.04	0
ZAF:natref	2008	—	13.716001529983394	—
ZAF:natref	2009	—	10.11640329174644	—
ZAF:natref	2010	—	10.569258156805544	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Supply • Refinery Utilisation

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_UTILISATION | 3/7

Units: fraction

Country / asset	Year(s)	Shared	High demand: raw alternatives	Low
ZAF:natref	2011	—	12.269760152634946	—
ZAF:natref	2012	—	13.550740237837728	—
ZAF:natref	2013	—	14.293444875066665	—
ZAF:natref	2014	—	14.529909583333335	—
ZAF:natref	2015	—	12.244166666666667	—
ZAF:natref	2016	—	11.653940311513589	—
ZAF:natref	2017	—	11.984234279716526	—
ZAF:natref	2018	—	12.926871155855082	—
ZAF:natref	2019	—	12.644779320295711	—
ZAF:natref	2020	—	11.439035777818907	—
ZAF:natref	2021	—	13.353394777286253	—
ZAF:natref	2022	—	0.873 [x2]; 2022 [x2]; 16.085106382978726	0.873
ZAF:natref	2023	—	0.875 [x2]; 2023 [x2]; 16.163120567375888	0.875
ZAF:natref	2024	—	0.88; 0.7; 2024 [x4]; 15.502793296089386	0.7
ZAF:natref	2025	—	0.85; 0.7; 2025 [x4]; 13.751592356687897	0.7
ZAF:natref	2026	—	0.85; 0.7; 2026 [x4]	0.7

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Supply • Refinery Utilisation

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_UTILISATION | 4/7

Units: fraction

Country / asset	Year(s)	Shared	High demand: raw alternatives	Low
ZAF:natref	2027	—	0.85; 0.7; 2027 [x4]	0.7
ZAF:natref	2028	—	0.85; 0.7; 2028 [x4]	0.7
ZAF:natref	2029	—	0.85; 0.7; 2029 [x4]	0.7
ZAF:natref	2030	—	0.85; 0.7; 2030 [x4]	0.7
ZAF:natref	2031	—	0.85; 0; 2031 [x4]	0
ZAF:natref	2032	—	0.85; 0; 2032 [x4]	0
ZAF:natref	2033	—	0.85; 0; 2033 [x4]	0
ZAF:natref	2034	—	0.85; 0; 2034 [x4]	0
ZAF:natref	2035	—	0.85; 0; 2035 [x4]	0
ZAF:natref	2036	—	0.85; 0; 2036 [x4]	0
ZAF:natref	2037	—	0.85; 0; 2037 [x4]	0
ZAF:natref	2038	—	0.85; 0; 2038 [x4]	0
ZAF:natref	2039	—	0.85; 0; 2039 [x4]	0
ZAF:natref	2040	—	0.85; 0; 2040 [x4]	0
ZAF:natref	2041	—	0.85; 0; 2041 [x4]	0
ZAF:natref	2042	—	0.85; 0; 2042 [x4]	0

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Supply • Refinery Utilisation

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_UTILISATION | 5/7

Units: fraction

Country / asset	Year(s)	Shared	High demand: raw alternatives	Low
ZAF:natref	2043	—	0.85; 0; 2043 [x4]	0
ZAF:natref	2044	—	0.85; 0; 2044 [x4]	0
ZAF:natref	2045	—	0.85; 0; 2045 [x4]	0
ZAF:natref	2046	—	0.85; 0; 2046 [x4]	0
ZAF:natref	2047	—	0.85; 0; 2047 [x4]	0
ZAF:natref	2048	—	0.85; 0; 2048 [x4]	0
ZAF:natref	2049	—	0.85; 0; 2049 [x4]	0
ZAF:natref	2050	—	0.85; 0; 2050 [x4]	0
ZAF:petrosa	2022–2050	—	0 [x2]	0
ZAF:sapref	2008	—	545824000; 0.64839110099925	—
ZAF:sapref	2009	—	486528000; 0.7196233473550281	—
ZAF:sapref	2010	—	516022000; 0.7708677259201796	—
ZAF:sapref	2011	—	558637000; 0.8037375257534161	—
ZAF:sapref	2012	—	559027000; 0.8438038414122238	—
ZAF:sapref	2013	—	568128000; 0.8912593694532472	—
ZAF:sapref	2014	—	608701000; 0.9425959091337544	—

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Supply • Refinery Utilisation

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_UTILISATION | 6/7

Units: fraction

Country / asset	Year(s)	Shared	High demand: raw alternatives	Low
ZAF:sapref	2015	—	584740000; 1	—
ZAF:sapref	2016	—	606365000; 1.06341	—
ZAF:sapref	2017	—	667908000; 1.1194836093	—
ZAF:sapref	2018	—	727979000; 1.17120375204966	—
ZAF:sapref	2019	—	707740000; 1.219574467009311	—
ZAF:sapref	2020	—	623654000; 1.2594301605911753	—
ZAF:sapref	2021	—	674535000; 1.3150214078796698	—
ZAF:sapref	2022	—	0 [x2]; 839218000; 1.41	0
ZAF:sapref	2023	—	0 [x2]; 837112000; 1.41	0
ZAF:sapref	2024	—	0 [x2]; 980518000; 1.432	0
ZAF:sapref	2025	—	0 [x2]; 970558422.9711272; 1.57	0
ZAF:sapref	2026–2050	—	0 [x2]	0
ZAF:sasol	2022	—	0.9; 0.8947; 1 [x2]	0.8947
ZAF:sasol	2023	—	0.9123; 0.9013; 1 [x2]	0.9013
ZAF:sasol	2024	—	0.9246; 0.9079; 1 [x2]; 34; 31578	0.9079
ZAF:sasol	2025	—	0.9369; 0.9145; 1 [x2]; 25; 29341	0.9145

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Supply • Refinery Utilisation

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_UTILISATION | 7/7

Units: fraction

Country / asset	Year(s)	Shared	High demand: raw alternatives	Low
ZAF:sasol	2026	—	0.9492; 0.9211; 1 [x2]; 84; 39587	0.9211
ZAF:sasol	2027	—	0.9615; 0.9277; 1 [x2]; 235; 41103	0.9277
ZAF:sasol	2028	—	0.9737; 0.9342; 1 [x2]; 323; 40743	0.9342
ZAF:sasol	2029–2030	—	0.9437; 0.8781; 1 [x2]; 323; 40743	0.8781
ZAF:sasol	2031–2032	—	0.9737; 0.8781; 1 [x2]; 323; 40743	0.8781
ZAF:sasol	2033	—	0.9446; 0.8444; 1 [x2]; 323; 40743	0.8444
ZAF:sasol	2034	—	0.9155; 0.8107; 1 [x2]; 323; 40743	0.8107
ZAF:sasol	2035	—	0.8863; 0.777; 1 [x2]; 323; 40743	0.777
ZAF:sasol	2036–2040	—	0.792; 0.777; 1 [x2]; 323; 40743	0.777
ZAF:sasol	2041–2050	—	0.792; 0; 1 [x2]; 323; 40743	0

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Supply • Refinery Availability

2026 draft | Domestic petrol, diesel and jet production | REG-SUPPLY-REFINERY_AVAILABILITY | 1/1

Units: fraction

Parameter / scenario	Coverage	Declared value
value	Shared	0.9

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Vehicles • Ev Scurve

2026 draft | Petrol and road diesel | REG-VEHICLES-EV_SCURVE | 1/1

Units: ratio (penetration), year (midpoint)

Parameter / scenario	Coverage	Declared value
high_demand.max_penetration	ZAF	0.3
high_demand.midpoint	ZAF	2040
high_demand.slope	ZAF	3
low_demand.max_penetration	ZAF	0.9
low_demand.midpoint	ZAF	2036
low_demand.slope	ZAF	2
high_demand.max_penetration	BWA, LSO, NAM, SWZ	Missing (null)
high_demand.midpoint	BWA, LSO, NAM, SWZ	Missing (null)
high_demand.slope	BWA, LSO, NAM, SWZ	Missing (null)
low_demand.max_penetration	BWA, LSO, NAM, SWZ	Missing (null)
low_demand.midpoint	BWA, LSO, NAM, SWZ	Missing (null)
low_demand.slope	BWA, LSO, NAM, SWZ	Missing (null)

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Vehicles • New Vehicle Regression

2026 draft | Petrol and road diesel | REG-VEHICLES-NEW_VEHICLE_REGRESSION | 1/1

Units: vehicles per year (response), ZAR-per-capita constant 2015 (driver)

Parameter / scenario	Coverage	Declared value
slope	ZAF	37.3149
intercept	ZAF	-2387744
r_squared	ZAF	0.511137
slope	BWA, LSO, NAM, SWZ	Missing (null)
intercept	BWA, LSO, NAM, SWZ	Missing (null)
r_squared	BWA, LSO, NAM, SWZ	Missing (null)

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Vehicles • Annual Km Per Vehicle

2026 draft | Petrol and road diesel | REG-VEHICLES-ANNUAL_KM_PER_VEHICLE | 1/1

Units: km per vehicle per year

Parameter / scenario	Coverage	Declared value
passenger	ZAF	17000
lcv	ZAF	28000
hcv	ZAF	70000
passenger	BWA, LSO, NAM, SWZ	Missing (null)
lcv	BWA, LSO, NAM, SWZ	Missing (null)
hcv	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Vehicles • Fuel Consumption

2026 draft | Petrol and road diesel | REG-VEHICLES-FUEL_CONSUMPTION | 1/1

Units: L / 100 km

Parameter / scenario	Coverage	Declared value
passenger.ice_petrol	ZAF	9.5
passenger.ice_diesel	ZAF	7.5
lcv.ice_petrol	ZAF	10.5
lcv.ice_diesel	ZAF	9.0
hcv.ice_petrol	ZAF, BWA, LSO, NAM, SWZ	Missing (null)
hcv.ice_diesel	ZAF	35.0
passenger.ice_petrol	BWA, LSO, NAM, SWZ	Missing (null)
passenger.ice_diesel	BWA, LSO, NAM, SWZ	Missing (null)
lcv.ice_petrol	BWA, LSO, NAM, SWZ	Missing (null)
lcv.ice_diesel	BWA, LSO, NAM, SWZ	Missing (null)
hcv.ice_diesel	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Vehicles • Petrol Diesel Split

2026 draft | Petrol and road diesel | REG-VEHICLES-PETROL_DIESEL_SPLIT | 1/1

Units: fraction (sums to 1 per country×segment)

Parameter / scenario	Coverage	Declared value
passenger.petrol	ZAF	0.83
passenger.diesel	ZAF	0.17
lcv.petrol	ZAF	0.3
lcv.diesel	ZAF	0.7
hcv.petrol	ZAF	0.0
hcv.diesel	ZAF	1.0
passenger.petrol	BWA, LSO, NAM, SWZ	Missing (null)
passenger.diesel	BWA, LSO, NAM, SWZ	Missing (null)
lcv.petrol	BWA, LSO, NAM, SWZ	Missing (null)
lcv.diesel	BWA, LSO, NAM, SWZ	Missing (null)
hcv.petrol	BWA, LSO, NAM, SWZ	Missing (null)
hcv.diesel	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Vehicles • Scrappage Rate

2026 draft | Petrol and road diesel | REG-VEHICLES-SCRAPPAGE_RATE | 1/1

Units: fraction per year

Parameter / scenario	Coverage	Declared value
passenger	ZAF	0.04
lcv	ZAF	0.05
hcv	ZAF	0.06
passenger	BWA, LSO, NAM, SWZ	Missing (null)
lcv	BWA, LSO, NAM, SWZ	Missing (null)
hcv	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Vehicles • New Vehicle Segment Split

2026 draft | Petrol and road diesel | REG-VEHICLES-NEW_VEHICLE_SEGMENT_SPLIT | 1/1

Units: fraction (sums to 1 per country)

Parameter / scenario	Coverage	Declared value
passenger	ZAF	0.78
lcv	ZAF	0.18
hcv	ZAF	0.04
passenger	BWA, LSO, NAM, SWZ	Missing (null)
lcv	BWA, LSO, NAM, SWZ	Missing (null)
hcv	BWA, LSO, NAM, SWZ	Missing (null)

Source date: Not recorded | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Missing regional values are declarations, not zeroes. Coverage combines only identical leaf/value entries.

Assumption inventory: Vehicles • Efficiency Improvement • Diesel

2026 draft | Petrol and road diesel | REG-VEHICLES-EFFICIENCY_IMPROVEMENT-DIESEL | 1/1

Units: fraction per year

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2020–2050	—	0.005	0.01

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Assumption inventory: Vehicles • Efficiency Improvement • Gasoline

2026 draft | Petrol and road diesel | REG-VEHICLES-EFFICIENCY_IMPROVEMENT-GASOLINE | 1/1

Units: fraction per year

Country / asset	Year(s)	Shared	High demand	Low demand
ZAF	2020–2050	—	0.01	0.015

Source date: 2025-03-24 | Owner: Nigel | Review: unreviewed | Confidence: unassessed

Year ranges repeat the exact value shown. [xN] retains duplicate records per year; — means no declaration.

Calculation conventions outside the input register

These choices also affect results. They are current code behaviour, not approved assumptions.

Convention	Current implementation / value	Fuel affected
Forecast horizon	2024–2050; inclusive annual/monthly dimensions	All products
Vehicle opening stock	Empty cohort book at 2005; fleet accumulated from regression-based additions	Petrol, road diesel
Vehicle efficiency	Cumulative improvement sets consumption of new cohorts; held fixed for cohort life	Petrol, road diesel
Missing / negative drivers	Vehicles use zero for missing GDP-year additions; vehicle and jet regression outputs clamped at zero	Petrol, diesel, jet
Held sector method	Industry, agriculture and marine scale baseline volume with GDP per capita and elasticity	Diesel, marine fuel oil
Power conversions	24 hours/day; 3600 MJ/MWh	Power diesel
Domestic supply conversions	365 days/year; 1000 bpd/kbpd	Petrol, diesel, jet
Duplicate production records	First non-null value per asset/year is selected; raw alternatives remain in the inventory	Domestic production
Balance and monthly allocation	Balance is demand less domestic production; core annual demand flat-allocated across 12 months	All current products

Source behaviour is documented [here](#); no code, registered assumption or calculation has been changed.